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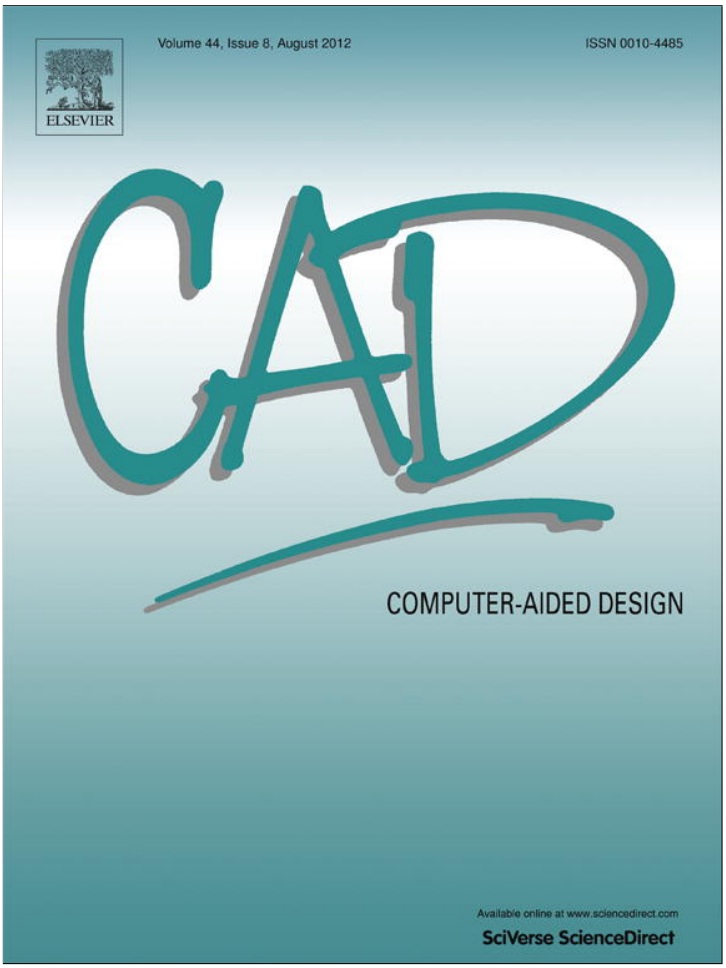

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Computational modeling of effects of intravascular stent design on key mechanical and hemodynamic behavior

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ABSTRACT

Atherosclerosis, a condition related to cholesterol build-up and thickening of the inner wall of the artery, narrows or occludes the artery lumen. A stent is a miniature medical device deployed in a stenotic artery to restore the blood flow. In this paper, we propose to apply the parametric design concept onto the stent design and integrate it with the developed FEA/CFD models to evaluate several key clinically-relevant functional attributes recommended by the FDA. These key clinical attributes include stresses/strains, fatigue resistance, radial strength, expansion recoil, and wall shear stresses, which have yet to be systematically investigated. Finite element models were developed to predict the mechanical integrity of a balloon-expandable stent at various stages such as crimping onto a balloon catheter, stent expansion, radial strength to resist blood vessels from collapsing, and service life in the human body when subjected to pulsatile blood pressure. Computational fluid dynamics models were developed to predict the wall shear stress distribution in stented arteries. A stent parametric analysis was conducted using the integrated computational schemes to systematically evaluate the effects of varying stent design parameters on key clinically-relevant functional attributes. Each stent design parameter was varied in its dimension from −30% to +30% (compared to the standard case) for sensitivity studies in attempts to find the most dominant design parameter for each key clinical attribute. The developed design/analytical schemes allow us to gain deeper insight into the fundamental stent issues and evaluate the mechanical/hemodynamic behavior of various stent designs.

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1. Introduction

Atherosclerosis is a condition in which plaque consisting of fatty substances such as cholesterol and triglyceride builds up and consequently narrows the artery lumen. A stent is a small, coiled wire-mesh tube that can be deployed into an artery and expanded percutaneously using a balloon catheter during angioplasty to open a narrowed artery. The expanded stent exerts radial force against the artery wall, thereby creating a scaffolding structure and preventing re-closure of the artery [1]. Today intravascular stenting has emerged as one of the standards for the treatment of vascular diseases. Over the past two decades, the field has seen numerous innovations in attempts to perfect the percutaneous management of vascular diseases [2–8]. With this revolutionary technology, stents today have extended their indications to most of the vasculature system in our body.

When making a stent, several procedures of design, manufacturing, and testing are typically employed. A 2D stent design is first sketched on the x–y plane and coded into the 3D cylindrical coordinate system by wrapping the 2D sketch around a target cylinder using the commercial CAD/CAM/CAE software. The coded stent geometry is then fed into the laser machine and the design pattern is cut onto the hypotubes. The manufactured stents are finally tested to evaluate whether the stent performance meets the initial design intent. If not, the entire process has to repeat continuously until satisfactory stent performance is obtained. Each design cycle could take days to weeks for completion, which is a time-consuming and inefficient process. That is the reason why, in recent years, finite element analysis (FEA) and computational fluid dynamics (CFD) have become increasingly important tools to obtain rapid evaluation of the stent acute performance and prediction of the stent long-term fatigue life. Such models could give insights into various aspects of the stent design that may consequently reduce the risk of vascular injury and restenosis. It also provides an extensive amount of information under highly controlled conditions, making it feasible to screen various stent design iterations prior to costly prototyping.

FEA analysis was first exploited to evaluate the stent deployment [9–11]. Dumoulin and Cochelin and Etave et al. modeled the stent deployment via a unit cell loaded with radial internal pressure. Donnelly et al. compared the stress/strain contour plots of different stent geometries via a unit cell model. Later on, various studies aimed to improve the modeling techniques and evaluate other stent properties were proposed [12–20]. For example,

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Gervaso et al. and Zahedmanesh et al. studied various modeling techniques for balloon-expandable stents. Holzapfel et al. defined several indicators based on stent–artery interaction to evaluate stent performance with different stent designs. Hsiao et al. evaluated the renal stent fatigue life in respiration based on Goodman life analysis.

In 2010, the FDA published an official document to serve as the guidance for the medical device industry and listed several key clinically-relevant functional attributes in which the FDA is interested when reviewing future new stent submissions. These key clinical attributes include stresses/strains, fatigue resistance, radial strength, and expansion recoil, which have yet to be investigated systematically with respect to the stent design parameters such as geometry and pattern. Therefore, in this paper, we propose to apply the parametric design concept onto the stent design and integrate it with the developed FEA and CFD models to evaluate these key clinical attributes as a function of the stent design parameters. With these new ideas employed, a modified stent drawing can be created in just a few seconds and stent design could be accomplished within much less time than with traditional approaches lacking parametric stent design and/or FEA/CFD analysis. A systematic stent parametric analysis was performed to assess the effects of varying stent design parameters on key clinically-relevant functional attributes recommended by the FDA using the developed computational design and analysis schemes.

It is important to develop this integrated CAD/FEA/CFD scheme for the stent industry, as this process could potentially reduce the product development cycle from months to weeks and save precious time for the industry in the future. The authors are currently working on integrating the parametric stent design and FEA/CFD analysis altogether by using the Graphical User Interface (GUI) so the entire process could be further automated and streamlined without any manual setting-ups in between.

2. Computational model development

2.1. Parametric stent design

The objective of parametric design is to reduce drafting time caused by having to make multiple corrections and additions to drawing files, which is common in regular CAD software such as AutoCAD. Parametric design is the latest development in CAD software and refers to the inclusion of parametric dimensions and variables embedded within 3D objects (i.e., parameters such as height, depth, thickness, and even attributes such as material). It is a method of linking dimensions and variables to geometry in such a way that when the defined parametric values change, the drawing changes as well. A parameter is a variable to which other variables are related, and these other variables can be obtained by means of parametric mathematical equations. It usually requires the designer to consider the consequences of his planned actions carefully prior to drawing. Therefore, it could take up to a week or two for an experienced stent designer to build the parametric relationships and debug the program for the very initial design. Parametric modification can be accomplished with a designed spreadsheet, script, or by manually changing dimension text in the model.

Parametric stent design has been previously discussed in the literature [21,22]. A typical stent has a series of nested rings interconnected with small bridging connectors. It is an assembly of a repeated unit cell, the smallest representative element, in a periodic pattern. Fig. 1 illustrates an intravascular stent and the definition of three representative parameters (crown radius, strut thickness, and strut width) in a single unit cell. Crown radius is one of the most critical parameters among all, as

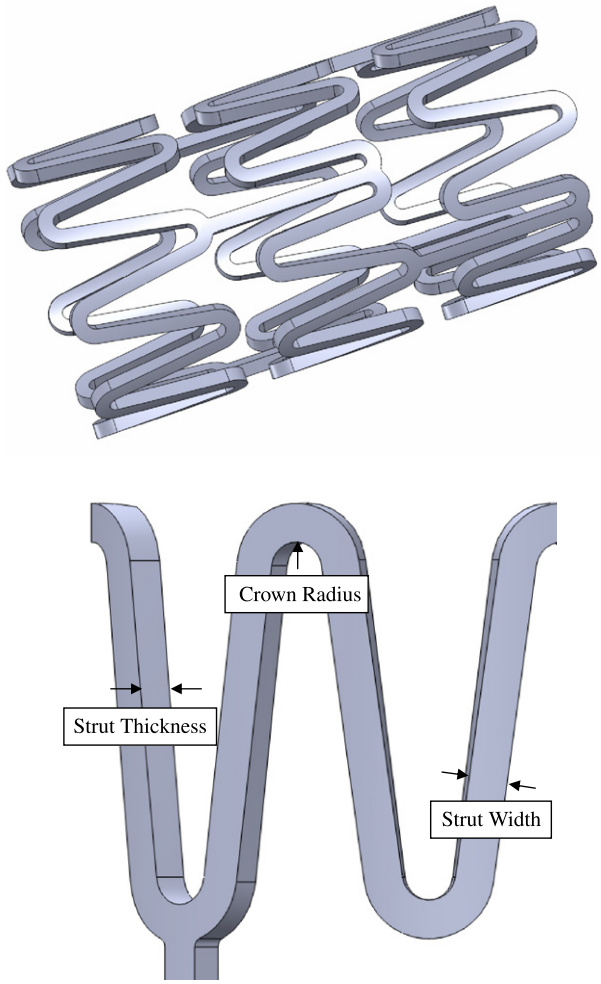

Fig. 1. Stent geometric parameter definition.

it controls many key stent functional attributes such as acute fracture, elastic recoil, stent retention on catheters, and crimp profile. Strut thickness is believed to play an important role in the restenosis process, whereas strut width dictates the stent radial strength. Both strut thickness and strut width also affect the final stent crimp profile. Our parametric stent design started with this single unit cell. All important stent geometries such as tubing diameter, strut width/thickness/length, crown radius/number, etc., were parameterized by mathematical equations. An example of the parametric equation used in this paper was:

Circumferential distance between crown centers

$$= pi * tubing\ diameter / crown\ number.$$

Some geometric constraints were defined to limit the changes of some variables. For example, the strut lines were constrained to be tangent to the crown arcs; two circles were constrained to be concentric to form a crown. A constraint is a relation that limits the behavior of an entity or a group of entities. If a model is under-constrained, it cannot be resolved because some additional parameters still have to be specified; if a model is over-constrained, it cannot be resolved because there is a contradiction somewhere. Constraint modeling requires all the defined constraints to be fulfilled and the degrees of freedom of the model reduced to zero.

After completing the unit cell sketch and building the mathematical equations, it can be copied multiple times circumferentially and axially to form a complete 2D stent drawing with embedded parametric relationships. The 2D stent drawing was wrapped onto the surface of a cylinder and the strut thickness was

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then protruded. A spreadsheet was generated with all the parameters listed on the table for easy parametric modifications. By changing the parameter values, design modifications and creation of a family of stent designs can be performed in remarkably quick time, usually within seconds, compared with the redrawing required by the traditional CAD method which could take hours per sketch for an experienced stent designer repeatedly. This fast parametric design scheme can be combined with the developed FEA/CFD models to become an integrated and powerful computational tool for future stent design. Our methodology differs from the existing work in the literature as we propose a coherent stent design process that incorporates all essential components including parametric design and FEA/CFD analysis.

2.2. Finite element analysis

The FDA published "Non-Clinical Engineering Tests and Recommended Labeling for Intravascular Stents and Associated Delivery Systems" on April 18, 2010 to list key clinically-relevant functional attributes the agency is interested, including stresses/strains, fatigue resistance, radial strength, and expansion recoil, for future consideration of new stent submissions [23]. In this paper, these key clinical attributes were used to assess the mechanical integrity of a balloon-expandable stent.

2.2.1. Finite element model

A stent placed in the vasculature is subjected to various modes of loading that may consequently compromise the mechanical integrity of the stent during their functional life. In this paper, several finite element models were developed to evaluate the stent mechanical integrity and fatigue resistance to various loading conditions consistent with the current practice. It includes crimping onto a balloon catheter, stent free expansion, their corresponding recoil, and pulsatile fatigue loading subjected to systolic/diastolic pressure:

Step 1: Stent crimping from 2.54 mm to 2 mm OD (crimp).
Step 2: Stent recoil after crimping (crimp recoil).
Step 3: Stent expansion to 7.0 mm ID (expansion).
Step 4: Stent recoil after expansion (expansion recoil).
Step 5: Stent fatigue under 180/80 mmHg systolic/diastolic pressure, or, stent radial strength.

After expanding to the target diameter, the stent experiences some elastic recovery as the balloon deflates. The stent size may become smaller than desired and result in a vessel wall apposition issue. The expansion recoil is the percentage difference in the stent diameter between post-balloon expansion and after balloon deflation:

$$\text{Expansion Recoil} = \frac{D_{\text{step3}} - D_{\text{step4}}}{D_{\text{step3}}}.$$

It should be noted that our stent expansion model was used to make relative comparisons of key clinical attributes among various stent design parameters. It is useful to differentiate a good stent design from bad ones and find the trends of their behavior; however, it does not simulate realistic in vivo conditions where stent interactions with angioplasty balloons and hyperelastic arteries have to be considered. Several research groups already demonstrated the importance and feasibility of including angioplasty balloons and hyperelastic arteries in their FEA models while replicating in vivo coronary conditions [12,16,17].

2.2.2. Stent geometry and meshing

The ABAQUS/Standard finite element solver (Dassault Systemes Simulia Corp., Providence, RI, USA) was used to perform the stent analysis. Since a stent has repeated patterns in the axial direction, three representative rings were modeled to save computational time. In order to simulate the crimping onto a balloon catheter and the stent free expansion steps, two cylinders with the diameters of 2.54 mm and 1.12 mm were added into the model, with one inside the stent and the other outside the stent.

Table 1
Performed mesh sensitivity analysis.

Element number	3 × 3	4 × 3	5 × 3	6 × 3	7 × 3	8×3
PEEQ (%)	47.0	47.7	47.7	47.7	47.4	47.4
RS (N/mm)	4.03	4.03	3.99	4.00	3.99	3.99

The stent model was meshed with the 8-node linear brick element in incompatible mode (C3D8I). The total element number was 82266 for each stent model. When performing FEA analysis, it was observed that the stress/strain variation across the strut width was much more severe than through the strut thickness and as a result, we decided to put more elements in the width direction. Mesh sensitivity analysis was conducted on PEEQ and radial strength for various element sizes starting from 3 × 3 per strut cross-section all the way to 8 × 3 (Table 1). Simulated results converged relatively quickly and their values became stabilized when the mesh size was greater than 3 × 3. Therefore, we selected a conservative 6 × 3 as the final element size to ensure that stress/strain variation was adequately captured and used it for all cases investigated. The inside and outside cylinders were meshed with the 4-node quadrilateral surface element (SFM3D4). The stent and two cylinders were meshed by the sweeping method.

2.2.3. Material properties and boundary conditions

The material properties of L-605 cobalt–chromium alloy were used along with the ABAQUS von Mises plasticity model with isotropic hardening. The Young's modulus, Poisson ratio, and the yield stress are 203 GPa, 0.3, and 590 MPa, respectively. The ultimate stress, ultimate strain, and fatigue endurance limit are 1689 MPa, 60%, and 483 MPa, respectively [19,20].

A frictionless contact was defined to prevent surface penetration with the following contact pairs considered: (1) surface contact between the inner surface of the outer cylinder and the outer surface of the stent, (2) surface contact between the outer surface of the inner cylinder and the inner surface of the stent, (3) side contact between any two stent struts during crimping.

2.2.4. Goodman fatigue life analysis

Failure of a stent due to fatigue may result in loss of a radial support of the stented vessel, thrombus formation or focal restenosis, or in perforation of the vessel by the broken stent struts. This fatigue analysis, combined with the previous stress/strain analysis and accelerated durability bench testing, provides an indication of chronic device durability. The FDA recommends using a Goodman life analysis to determine the fatigue safety factor of the stent to physiologic loading up to 4×10^8 fatigue cycles.

Following the stent deployment, systolic and diastolic arterial blood pressures (180/80 mmHg) were applied to simulate the pulsatile fatigue loading [19,20]. To account for the loading imposed by the arterial wall, an arterial pressure loading corresponding to the interaction between the stent and the artery was imposed on the stent. The Goodman life analysis, as recommended by the FDA, was performed using the multi-axial stress state experienced during pulsatile loading to determine the fatigue resistance of a stent. The Goodman relation states that fatigue failure will occur if the stress state in the component satisfies the relation:

$$\left(\frac{\sigma_a}{\sigma_e}\right) + \left(\frac{\sigma_m}{\sigma_u}\right) \geq 1$$

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where σ_a is the stress amplitude applied to the component, σ_e is the modified material endurance limit for non-zero mean stress, σ_m is the mean stress applied to the component, and σ_u is the material ultimate stress. The Goodman fatigue analysis was performed using the following effective mean stress and effective stress amplitude equations [24]:

$$\sigma_m = \frac{1}{\sqrt{2}}\sqrt{(\sigma_{1m}-\sigma_{2m})^2+(\sigma_{2m}-\sigma_{3m})^2+(\sigma_{3m}-\sigma_{1m})^2}$$
$$\sigma_a = \frac{1}{\sqrt{2}}\sqrt{(\sigma_{1a}-\sigma_{2a})^2+(\sigma_{2a}-\sigma_{3a})^2+(\sigma_{3a}-\sigma_{1a})^2}$$

where σ_m is the effective mean stress, σ_a is the effective stress amplitude, $\sigma_{1m}, \sigma_{2m}, \sigma_{3m}$ are the principal mean stresses, and $\sigma_{1a}, \sigma_{2a}, \sigma_{3a}$ are the principal stress amplitudes.

The principal stresses $\sigma_1, \sigma_2, \sigma_3$ were first gathered at each integration points for the pulsatile loading. These principal stresses were used to calculate the principal mean stresses and stress amplitudes. The principal mean stress is half of the sum of the principal stresses in the same direction at systolic/diastolic, whereas the principal stress amplitude is half of the difference of the principal stresses in the same direction at systolic/diastolic:

$$\sigma_{im} = \frac{\sigma_{i(\text{systolic})}+\sigma_{i(\text{diastolic})}}{2} \quad (i = 1, 2, 3)$$
$$\sigma_{ia} = \frac{\sigma_{i(\text{systolic})}-\sigma_{i(\text{diastolic})}}{2} \quad (i = 1, 2, 3).$$

After principal mean stresses and stress amplitudes were determined, the effective mean stress and the effective stress amplitude were then calculated at each integration point.

The Goodman diagram is a plot of the normalized stress amplitude σ_a/σ_e (on the y-axis) versus the normalized mean stress σ_m/σ_u (on the x-axis). The equation $(\sigma_a/\sigma_e) + (\sigma_m/\sigma_u) = 1$ represents the failure line on the Goodman diagram for non-zero mean stress with the modified endurance limit defined as:

$$\sigma_e = \sigma_{e0}\left(1-\frac{\sigma_m}{\sigma_u}\right)$$

where σ_{e0} is the material endurance limit for zero mean stress. Data points closer to the failure line on the Goodman diagram indicate that specific locations of a stent associated with those data points are prone to fatigue failure.

In order to quantify the likelihood of fatigue failure for a stent, the Fatigue Safety Factor (FSF) is defined as the ratio of the stress amplitude against the modified endurance limit. It quantifies the proximity of the mean stress and stress amplitude at the given integration points to the Goodman failure line. FSF less than 1.0 indicates the fatigue failure.

$$\text{FSF} = \frac{\sigma_e}{\sigma_a}.$$

2.2.5. Stent radial strength

Radial strength characterizes the ability of the stent to resist collapse under short-term or long-term external loads from the artery wall. It is one of the most important properties of the stent. To calculate the stent radial strength, another outer cylinder of diameter 10 mm was added to the outside of the stent to simulate the radial strength testing. This cylinder was meshed with surface element (SFM3D4). The cylinder was compressed in a fashion similar to that in step 1 to force the stent to crimp down to a smaller size. The reaction forces in the radial direction induced on the cylinder nodes were calculated at each time stamp. Hoop strength is defined as the maximum value of the total reaction forces during crimping. Stent radial strength is defined as the hoop strength divided by the total stent length.

2.3. Computational fluid dynamics

2.3.1. Governing equations

The flow is governed by the conservation of mass and momentum, given by the following flow equations, respectively:

$$\nabla \cdot v = 0$$
$$\rho\left(\frac{\partial v}{\partial t} + v \cdot \nabla v\right) = -\nabla p + \mu \nabla^2 v + f.$$

The non-Newtonian nature of blood can be modeled using the Carreau model:

$$\mu = \mu_\infty + (\mu_0 - \mu_\infty)\left[1 + (\lambda\dot{\gamma})^2\right]^{(n-1)/2}$$

where μ_∞ and μ_0 are viscosities when shear rate becomes infinity and zero, respectively, $\dot{\gamma}$ is shear rate, and λ and n are material coefficients. These parameters can be obtained by fitting the Carreau model to the experimental data where the fitted values are $\mu_\infty = 0.0035$ kg/m s, $\mu_0 = 0.25$ kg/m s, $\lambda = 25.00$ s, and $n = 0.25$ [25,26].

As aforementioned, the Wall Shear Stress (WSS) is strongly related to the restenosis process and used as the major indicator in this paper. The magnitude of WSS vector is calculated as:

$$\tau_w = \mu\left[2\left(\frac{\partial u}{\partial x}\right)^2 + 2\left(\frac{\partial v}{\partial y}\right)^2 + 2\left(\frac{\partial w}{\partial z}\right)^2 + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right)^2\right]^{\frac{1}{2}}$$

where u, v, and w are the x, y and z components of the velocity vector, respectively.

2.3.2. Material properties and boundary conditions

The blood flow in a stented artery was assumed to be steady, non-Newtonian, incompressible, and laminar. The governing equations were solved by the commercial CFD solver FLUENT (ANSYS, Inc., Canonsburg, PA, USA) using its pressure–velocity coupled algorithm. This algorithm is well-established and able to solve the velocity and pressure fields simultaneously to provide robust solutions. The constructed fluid domain was illustrated in Fig. 2 with the inlet velocity profile provided. The stent was placed in a uniform-sized pseudo artery. The artery inner diameter was assumed the same as the stent outer diameter after expansion recoil from the previous FEA analysis. Depending on the degrees of the stent expansion recoil, each CFD model had slightly different artery diameters in attempts to reflect more realistic clinical conditions. A fully-developed and steady blood flow of 1000 ml/min was prescribed as the inlet boundary condition, while uniform and fixed pressure was prescribed as the outlet boundary condition. The artery wall was assumed to be fixed and no-slip. Blood was modeled as an incompressible non-Newtonian fluid with a density of $\rho = 1060$ kg/m^3. Each stented artery had approximately 8,000,000 unstructured mesh grids.

2.3.3. Wall Shear Stress (WSS)

After stenting, restenosis may occur if excessive proliferation of neo-intimal tissues blocks the blood flow again. Although restenosis is a complicated physiological phenomenon, a growing number of experts suspect that it may be due to altered wall shear stress distribution in stented arteries. Furthermore, several clinical studies reported that low WSS (less than 5 dynes/cm^2) or oscillating WSS was observed at the location where neo-intimal thickening in stented arteries is the greatest [27–30].

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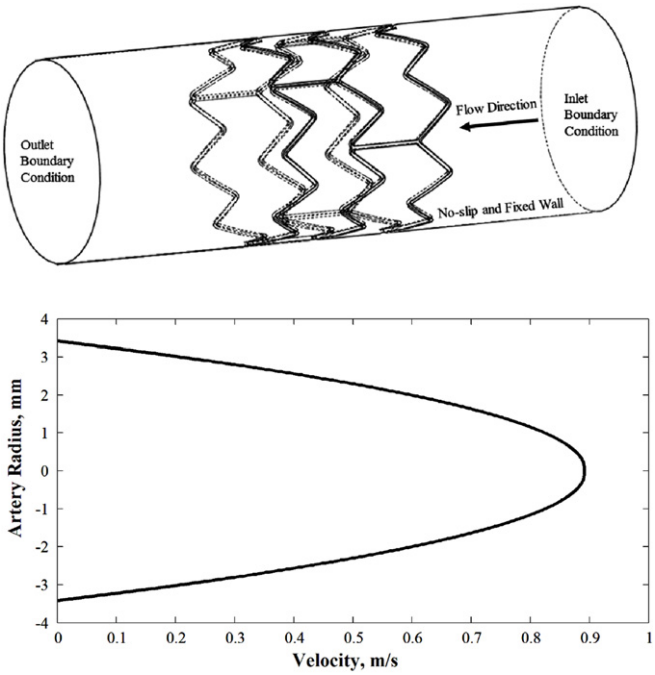

Fig. 2. Constructed fluid domain and inlet velocity profile.

Table 2
Various stent strut geometries used in stent parametric analysis.

Crown sensitivity study		Strut thickness sensitivity study		Strut width sensitivity study	
Case 1	CR + 30% CR	Case 1	ST + 30% ST	Case 1	SW + 30% SW
Case 2	CR + 15% CR	Case 2	ST + 15% ST	Case 2	SW + 15% SW
Case 3	Standard	Case 3	Standard	Case 3	Standard
Case 4	CR − 15% CR	Case 4	ST − 15% ST	Case 4	SW − 15% SW
Case 5	CR − 30% CR	Case 5	ST − 30% ST	Case 5	SW − 30% SW

3. Results and discussion

A stent parametric analysis was conducted using the developed parametric stent design and FEA/CFD analyses to systematically assess the effects of varying stent design parameters on key clinically-relevant functional attributes aforementioned. The selected stent geometries (or design parameters) for investigation included (1) crown radius (CR), (2) strut thickness (ST), and (3) strut width (SW) as defined in Fig. 1. Each parameter was varied in its dimension from −30% to +30% (compared to the standard case) for sensitivity studies between the stent parameters and key clinical attributes (Table 2).

3.1. Standard case

The simulation was first performed on a standard cobalt–chromium stent to establish the baseline information for key clinical attributes such as Equivalent Plastic Strain (PEEQ), Fatigue Safety Factor (FSF), Radial Strength (RS), Expansion Recoil (ER), and Wall Shear Stress (WSS). Their respective values were 47.7% (PEEQ), 2.56 (FSF), 4.00 N/mm (RS), 2.37% (ER), and 4.33 mm^2 (WSS).

Fig. 3 shows the contour plot comparison of the equivalent plastic strain for the standard stent at crimp and after crimp recoil. The maximum equivalent plastic strain occurred at the inner surface of the most highly curved region of the stent, while there were little to no stresses/strains on the straight portion of the stent. It is interesting to note that, after crimp recoil, the maximum equivalent plastic strain was unchanged. Fig. 4 shows the contour plot comparison of the equivalent plastic strain for the standard stent at expansion and after expansion recoil, while Fig. 5 shows the comparison of von Mises stress (unit: MPa) for the same stent at the same steps. It is shown that, after expansion recoil, the maximum von Mises stress dropped significantly to almost half of its original value, indicating energy release and stress relief, while the maximum equivalent plastic strain remained unchanged.

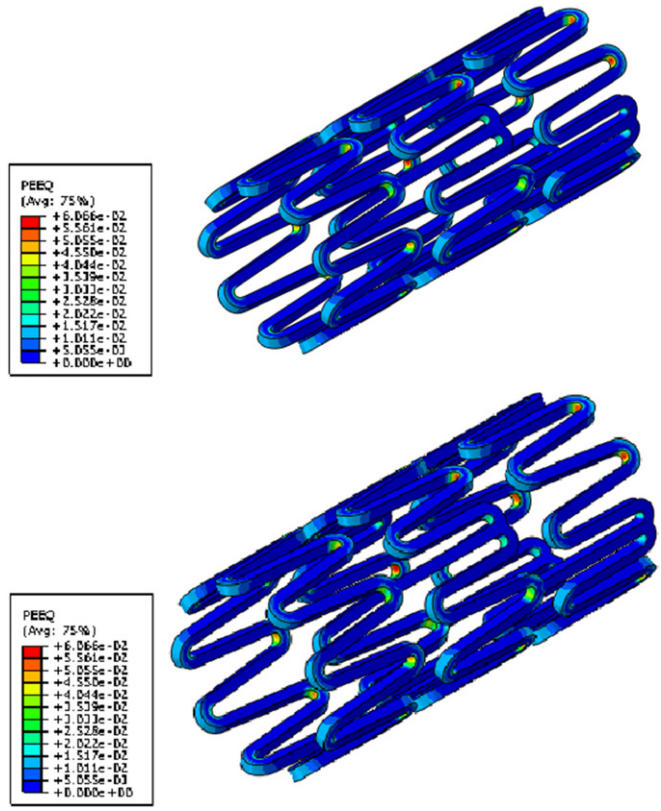

Fig. 3. Contour plot comparison of equivalent plastic strain for standard stent at crimp (top) and after crimp recoil (bottom).

Fig. 6 shows a Goodman diagram for the standard stent under systolic/diastolic blood pressure with the fatigue safety factor of 2.56. Calculated data were well below the Goodman diagram failure line, indicating that the standard stent is able to pass the fatigue life of 4×10^8 cycles under pulsatile fatigue loading. Fig. 7 shows the radial strength force–diameter curve when crimping down a stent from its expanded size to smaller diameters. In a clinical situation, this simulates the force exerted by the expanded stent against the artery wall to prevent it from collapsing inwards. As shown in the figure, the stent radial strength started to pick up force when the crimping initiated at the stent outer diameter of 7.2 mm and continued to rise. After reaching the maximum value of 4 N/mm at approximately 7.0 mm, the radial strength dropped perpetually. In this case, the stent eventually lost its lateral support as a scaffolding structure if the deformation was greater than 0.2 mm.

3.2. Stent parametric analysis

Each parameter was varied in its dimension from −30% to +30% for sensitivity studies of the stent parameters and key clinical attributes. The −30% to +30% variations in stent dimension represent the typical maximum stent dimensional tolerance allowed during stent laser-cutting and electro-polishing processes in the early days of bare metal stents. The degree and precision of laser-cutting and electro-polishing change the final dimension and tolerance of the stent struts (both width and

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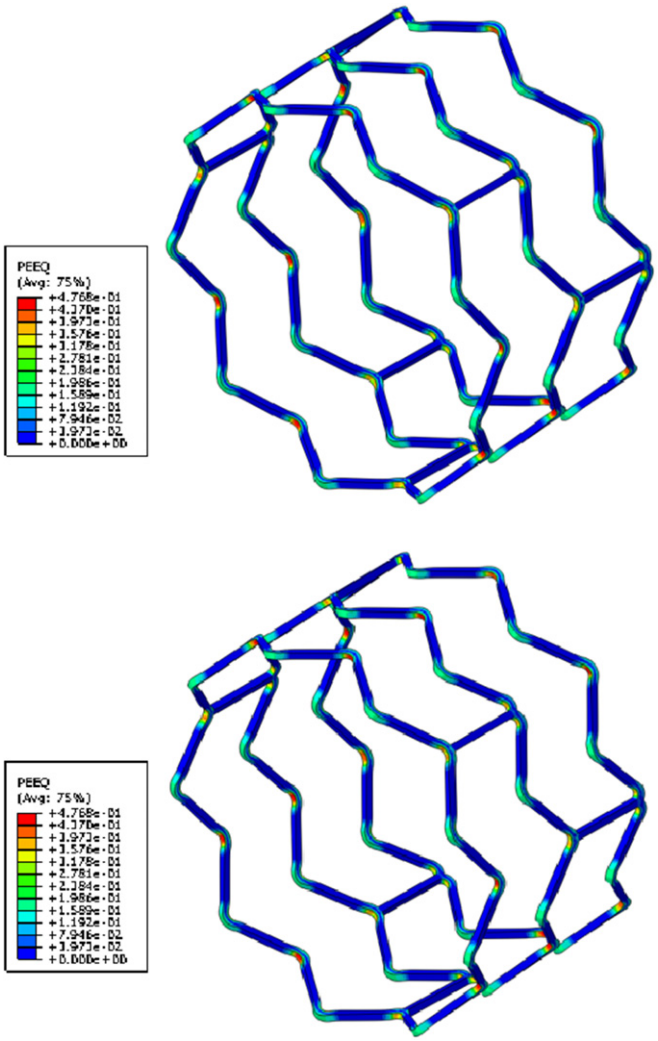

Fig. 4. Contour plot comparison of equivalent plastic strain for standard stent at expansion (top) and after expansion recoil (bottom).

Table 3
Simulation results of crown radius sensitivity study.

Key clinical attribute	CR − 30%	CR − 15%	Standard	CR + 15%	CR + 30%
PEEQ (%)	>60.0	55.1	47.7	41.4	36.6
FSF	1.60	2.35	2.56	2.72	2.85
RS (N/mm)	4.61	4.27	4.00	3.75	3.52
ER (%)	1.80	2.12	2.37	2.60	2.83
Low WSS area (mm^2)	12.06	11.91	11.85	11.81	11.80

thickness) and, therefore, significantly influence the overall stent mechanical and hemodynamic behavior. Tables 3–5 summarize the simulation results of three sensitivity studies for the crown radius, strut thickness, and strut width, respectively. These results were analyzed in terms of key clinical attributes in the subsequent discussions to help find the most dominant design parameter for each clinical attribute. They also demonstrated that laser-cutting and electro-polishing strongly influences the stent behavior and thus should be controlled precisely during stent manufacturing. It is the current industrial practice to control the stent dimensional tolerance to a much tighter degree.

3.2.1. Equivalent Plastic Strain (PEEQ)

Fig. 8 shows the simulation results of the equivalent plastic strain with respect to the crown radius, strut thickness, and strut width. It is noted that, in this paper, the crown radius was defined

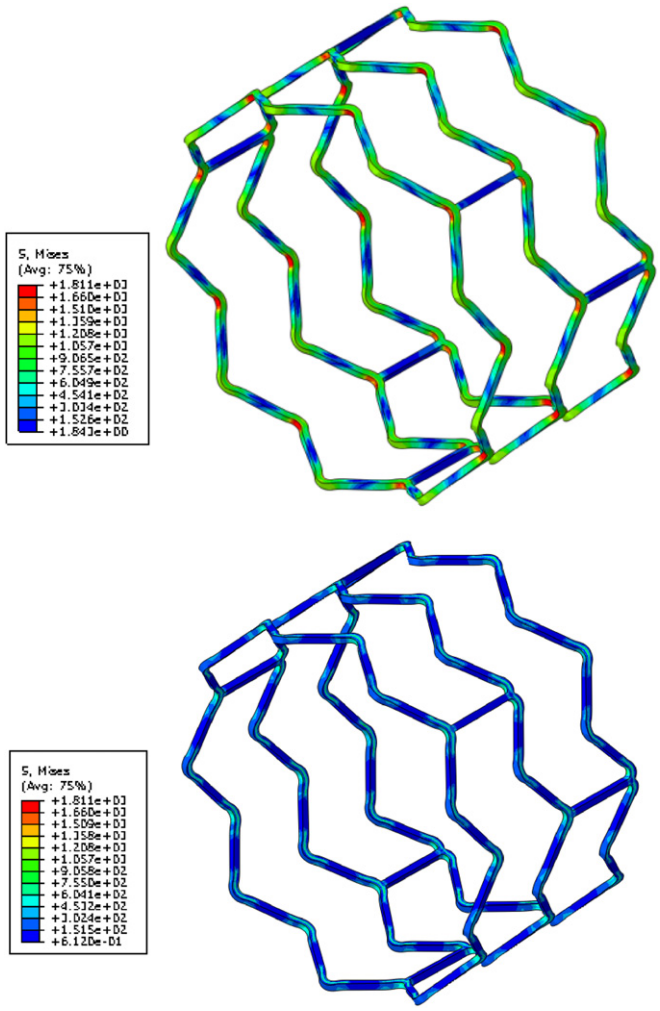

Fig. 5. Contour plot comparison of von Mises stress (unit: MPa) for standard stent at expansion (top) and after expansion recoil (bottom).

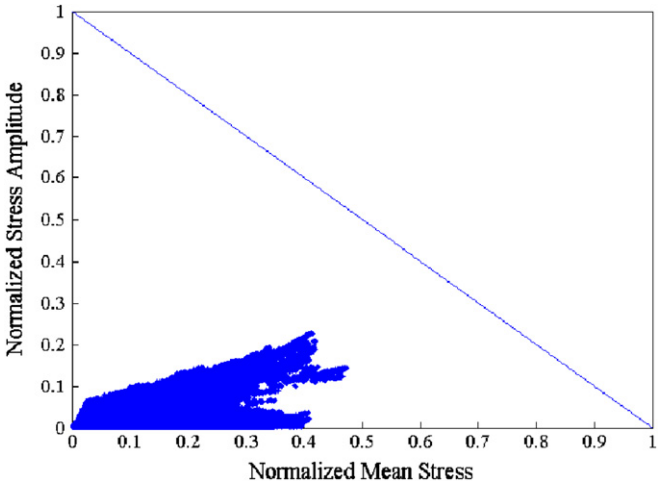

Fig. 6. Goodman diagram for standard stent under systolic/diastolic blood pressure.

as the inner radius of the crown and the strut width was defined as the difference between the fixed crown inner radius and a varied crown outer radius. This plot shows that reducing the crown radius or increasing the strut width resulted in higher equivalent plastic strain, while varying the strut thickness had very little

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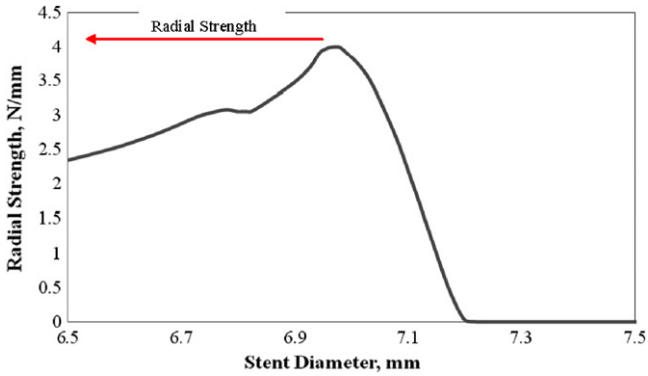

Fig. 7. Radial strength force–diameter curve.

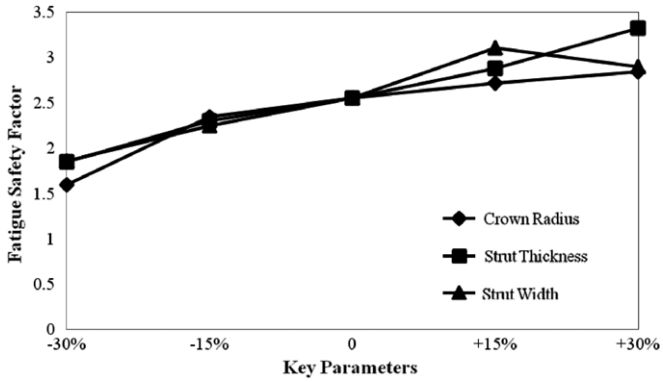

Fig. 9. Fatigue safety factor as a function of stent design parameters.

Table 4
Simulation results of strut thickness sensitivity study.

Key clinical attribute	ST − 30%	ST − 15%	Standard	ST + 15%	ST + 30%
PEEQ (%)	46.3	47.0	47.7	48.4	48.8
FSF	1.85	2.30	2.56	2.88	3.32
RS (N/mm)	2.66	3.35	4.00	4.61	5.22
ER (%)	2.47	2.41	2.37	2.35	2.29
Low WSS area (mm^2)	7.89	10.10	11.85	13.91	16.54

Table 5
Simulation results of strut width sensitivity study.

Key clinical attribute	SW − 30%	SW − 15%	Standard	SW + 15%	SW + 30%
PEEQ (%)	37.4	41.4	47.7	52.2	57.5
FSF	1.86	2.25	2.56	3.11	2.90
RS (N/mm)	1.912	2.878	4.00	5.11	6.44
ER (%)	2.45	2.40	2.37	2.36	2.35
Low WSS area (mm^2)	11.70	11.79	11.85	11.93	11.96

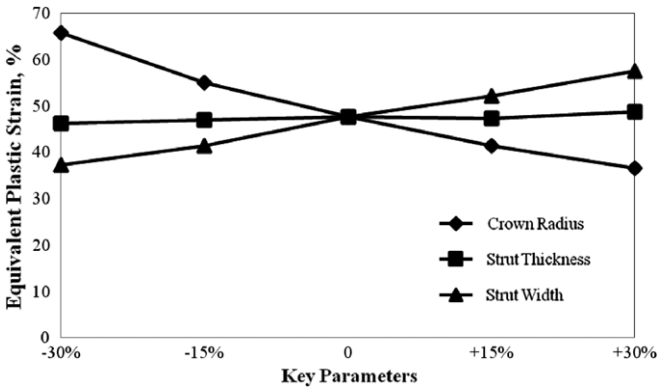

Fig. 8. Equivalent plastic strain as a function of stent design parameters.

effect. The equivalent plastic strain was more sensitive to the crown radius than the other two design parameters. Since the stent struts were loaded predominantly in bending, those struts connected to a larger crown radius were easier to bend, thereby inducing less plastic strain during deformation. A 30% increase in the crown radius reduced the maximum equivalent plastic strain by approximately 23% (from 47.7% strain to 36.6% strain).

A good stent design should be able to achieve a target plastic strain high enough to minimize elastic recoil and provide sufficient radial strength, while not exceeding the material ultimate strain. This suggests that the crown radius has to be deliberately designed, as it typically controls the maximum strain and several other properties. For the case of the crown radius with 30% reduction, the maximum equivalent plastic strain exceeded the material ultimate strain, suggesting that a stent with very small crown radius might lead to fracture issues during deployment.

3.2.2. Fatigue Safety Factor (FSF)

Fig. 9 shows the simulation results of the fatigue safety factor with respect to the crown radius, strut thickness, and strut width. In general, the fatigue safety factor trended higher as the stent design parameters increased their geometric dimensions. Thicker/wider stent struts (meaning a bulkier stent) or larger crown radii would improve the stent fatigue resistance against cyclic loading. A 30% increase in the strut thickness improved the fatigue safety factor by approximately 30% (from 2.56 to 3.32).

Based on these fatigue results, one would imagine a bulkier stent design may significantly increase the stent service life and thus is the direction to pursue. However, there are several adverse effects associated with the bulkier stent design. For example, thicker/wider stent struts or larger crown radii typically increase the overall stent profile (size). This is contradictory to the clinical request of a lower stent profile, which would allow access to smaller lesion sites and offer physicians more treatment options, easier operation, and greater delivery control. In addition, thicker struts result in lower wall shear stresses and larger recirculation zones at the downstream or backside of each stent struts. It is believed that low shear stress, particularly when its value is less than 5 dyne/cm^2, may lead to endothelial proliferation of smooth muscle cells, called restenosis.

3.2.3. Radial Strength (RS)

Fig. 10 shows the simulation results of the radial strength with respect to the crown radius, strut thickness, and strut width. This plot shows that increasing the strut thickness or strut width resulted in higher radial strength, while varying the crown radius had the opposite trend. This is not surprising considering the fact that a bulkier stent with thicker or wider struts could have stronger capability to resist collapse under external loads from the artery wall.

It is interesting to note that the radial strength is the clinical attribute most sensitive to the stent geometric dimensions. An increase of 30% in the strut width significantly boosted the radial strength by 61% (from 4.00 N/mm to 6.44 N/mm), while the same amount of increase in the strut thickness produced a smaller but still noticeable increase of 31% (from 4.00 N/mm to 5.22 N/mm). This suggests that the stent dimensional tolerance strongly influences the radial strength and should be controlled precisely during stent manufacturing in order to achieve the best clinical outcome.

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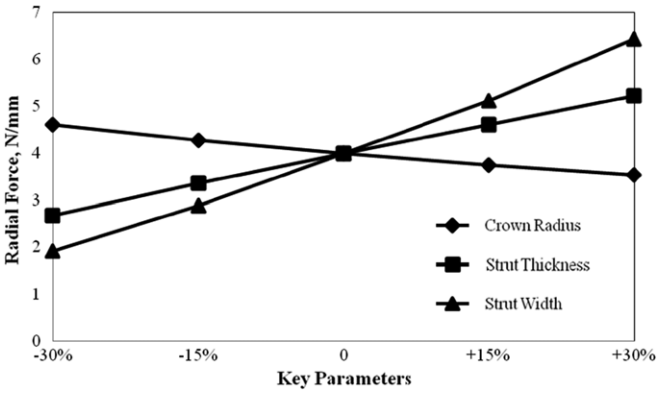

Fig. 10. Radial strength as a function of stent design parameters.

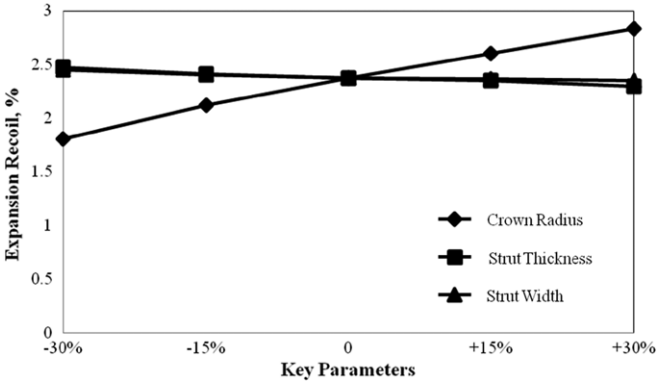

Fig. 11. Expansion recoil as a function of stent design parameters.

3.2.4. Expansion Recoil (ER)

Fig. 11 shows the simulation results of the expansion recoil with respect to the crown radius, strut thickness, and strut width. This plot shows that increasing the crown radius resulted in higher expansion recoil, while varying the strut width or thickness had very little effect. In general, the expansion recoil followed the opposite trends but with smaller margins when compared to the equivalent plastic strain plot. This might be attributed to the fact that the lower the equivalent plastic strain is, the higher the portion of the elastic strain is accounted for in the total strain. During recoil when the elastic deformation is recovered, the stent with a higher portion of elastic strain recoils more. A 30% increase in the crown radius increased the expansion recoil by approximately 19% (from 2.37% recoil to 2.83% recoil). In any cases, the recoil degree was relatively low, usually less than 3% for this specific design, for any stent dimensional changes and thus will not put this stent at a significant recoil risk.

3.2.5. Wall Shear Stress (WSS)

Fig. 12 shows the simulation results of the low WSS area with respect to the crown radius, strut thickness, and strut width. This plot shows that increasing the strut thickness resulted in much greater areas exposed to low WSS, while varying the strut width or crown radius had very little effect. Wall shear stress is very sensitive to the strut thickness.

Fig. 13 shows the contour plot comparison of the wall shear stress distributions for the stent with 30% reduction (top) and 30% increase (bottom) in the strut thickness. The blue color indicates the locations of the low wall shear stresses. It is apparent that the minimum WSS occurred at the recirculation zones located at the downstream or backside of each stent strut or joint of the strut. In order to quantity the effects, the total wall surface area exposed to low shear stress of 5 dyne/cm^2, below which restenosis

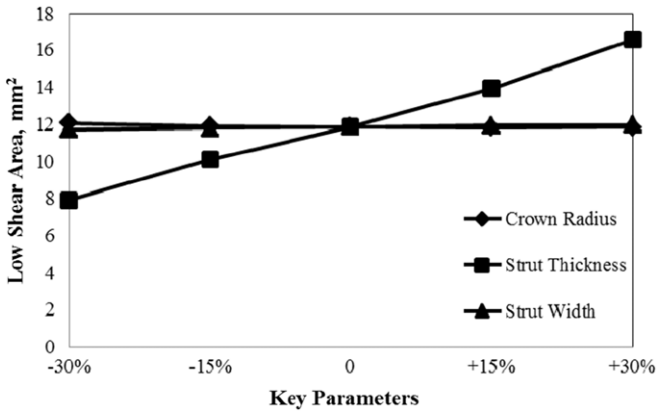

Fig. 12. Low wall shear stress area as a function of stent design parameters.

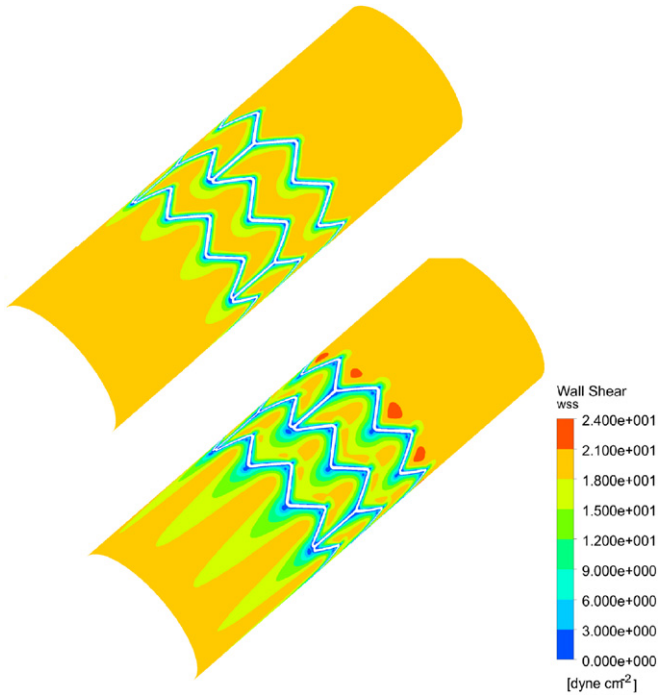

Fig. 13. Contour plots of wall shear stress for 30% reduction (top) and 30% increase (bottom) in strut thickness.

is likely to occur, was calculated and measured. At the flow rate of 1000 ml/min, the low WSS area was 7.89 mm^2 and 16.54 mm^2 for the 30% reduction and 30% increase in the strut thickness, respectively. This suggests that thinner stent design in attempts to improve the stent hemodynamic behavior could become one of the future directions of stent development in the medical device world.

3.3. Limitations

1. Our stent expansion model was used to make relative comparisons of key clinical attributes among various stent design parameters. It is useful to differentiate a good stent design from bad ones and find the trends of their mechanical and hemodynamic behavior; however, it does not simulate realistic *in vivo* conditions where stent interactions with angioplasty balloons and hyperelastic arteries have to be considered.
2. Since stent interactions with hyperelastic arteries were not simulated, our model was not able to predict an important

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clinical attribute called vessel injury score. However, as of today, the FDA prefers to review the real animal data than the simulated results for the assessment of vessel injury.

3. Our sensitivity studies allow us to compare the effects of varying single stent design parameters on key clinical attributes. However, it should be noted that the stent design optimization cannot be solely achieved by independent sensitivity analyses of single variables. The interactions among multiple variables have to be considered in order to optimize the stent design.

4. Conclusions

In this paper, we applied the parametric design concept to stent design and integrated it with the developed FEA/CFD models to evaluate several key clinical attributes as a function of the stent design parameters. It was demonstrated that the integrated computational schemes are able to help design and analyze a new stent quickly and significantly shorten the stent development cycle. A systematic stent parametric analysis was performed to assess the effects of varying stent design parameters on key clinical attributes. The results can be summarized as follows:

i. The most dominant stent design parameter for each key clinical attribute was identified: Crown radius is the most critical parameter for the equivalent plastic strain and the expansion recoil; strut width is the most critical parameter for the radial strength; and strut thickness is the most critical parameter for the fatigue safety factor and the wall shear stress. These conclusions are very important to future stent design.
ii. Among all key clinical attributes investigated in this paper, the radial strength is the most sensitive to the stent geometric dimensions. An increase of 30% in the strut width significantly boosted the radial strength by 61%, suggesting that the stent dimensional tolerance strongly influences the radial strength and should be controlled precisely during stent manufacturing in order to achieve the best clinical outcome.
iii. Along with the bioabsorbable stent, the thinner stent design for improving the stent hemodynamic behavior could become one of the future directions of stent development in the medical device world.

Acknowledgments

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